
Experimental Process

Clean-data evaluation, reproducible execution, and traceable result consolidation

The experiment compared **time-series foundation models** with **established forecasting baselines** under one consistent clean-data evaluation pipeline.

12 datasets

Rolling-origin evaluation

Four consolidated artifacts

Dataset portfolio: 12 clean evaluation datasets

Chronos / Monash datasets

- ▶ M1 Yearly
- ▶ M4 Hourly
- ▶ Weather
- ▶ M3 Quarterly
- ▶ Car Parts
- ▶ NN5 Weekly

External datasets

- ▶ National Illness
- ▶ Traffic
- ▶ PSM
- ▶ Elexon Electricity Demand
- ▶ USGS Streamflow
- ▶ BLS Macroeconomic Indicators

Six benchmark datasets + six external datasets

External-data provenance and reproducibility

Credential-dependent sources

- ▶ EIA-930 required an API credential.
- ▶ FRED-MD required an API credential.
- ▶ Both were excluded from the clean benchmark.

Keyless official sources

- ▶ Elexon demand was added as a replacement.
- ▶ BLS indicators were added as a replacement.
- ▶ USGS streamflow was retained as a credential-free source.

Official external observations were normalized and stored as **reproducible local snapshots**. Subsequent runs reused the same timestamped values and verified snapshot integrity.

Original clean model comparison

Three foundation models

- ▶ Amazon Chronos T5 Tiny
- ▶ Google TimesFM 2.5 200M
- ▶ Salesforce Moirai 2.0 Small

Five comparison models

- ▶ ARIMA(2,1,2)
- ▶ AutoARIMA
- ▶ Prophet
- ▶ DeepAR
- ▶ Seasonal Naive

Common tasks, forecast horizons, quantiles, and metric implementation

Separate Chronos checkpoint-scale experiment

Checkpoint	Parameters
Chronos T5 Mini	~20M
Chronos T5 Small	~46M
Chronos T5 Base	~200M
Chronos T5 Large	~710M

Execution settings

- ▶ All 12 datasets
- ▶ GPU execution
- ▶ bfloat16 inference
- ▶ Three repetitions per stochastic forecast
- ▶ 100 forecast samples per repetition

All 12 datasets completed with zero failed model tasks

Evaluation procedure: tasks and forecast outputs

Task construction

- ▶ Maximum historical context: 512 observations
- ▶ Dataset-specific horizons and seasonalities
- ▶ Multiple rolling forecast origins
- ▶ Generally three origins per selected series
- ▶ Two origins for Car Parts
- ▶ Task totals varied with eligible series and origins

Forecast representation

- ▶ Median used as the point forecast
- ▶ Quantiles from 0.1 through 0.9
- ▶ MAE, RMSE, sMAPE, MASE, and weighted quantile loss
- ▶ Rain occurrence error where applicable
- ▶ Positive-value MAE where applicable

Evaluation procedure: reproducibility record



Forecast horizon

Actual observations, point and mean forecasts, and all requested quantiles were retained at every timestamp.

Historical context

Pre-cutoff observations were retained so trajectories can be plotted without rerunning any model.

Three clean run groups

keyless_clean
main foundation-model
and baseline comparison

external_keyless_clean
dedicated six-external-
dataset run

heavier_chronos_clean
four Chronos sizes
across all 12 datasets

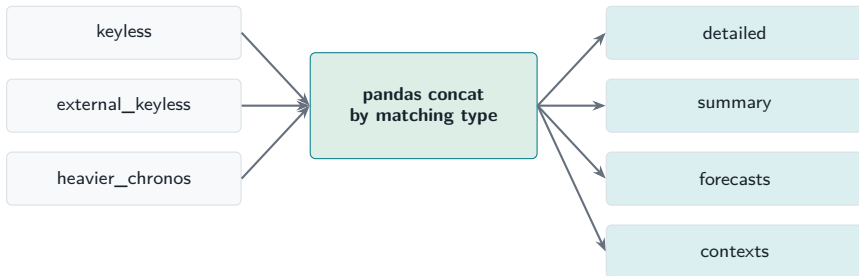
Robustness experiment

Stopped by request and not included in the final clean results.

Forecast audit

Maintained separately and excluded from final consolidation.

Final result consolidation



- ▶ Matching CSV types were concatenated with pandas.
- ▶ Artifact types remained separate because they represent different experimental levels.
- ▶ Every final row received a `source_run` value.
- ▶ Rows were preserved without deduplication so overlapping runs remain traceable.
- ▶ Final row counts were checked against all contributing source CSVs.

Consolidated artifacts: task- and dataset-level records

Detailed — 10,502 rows

One record per model repetition and forecasting task:

- ▶ dataset, series, origin, and cutoff
- ▶ model, repetition, and random seed
- ▶ forecast-distribution type and metrics
- ▶ duration, diagnostic notes, and errors

`final_clean_detailed.csv`

Summary — 176 rows

One aggregate record per dataset-model combination:

- ▶ aggregate metric fields
- ▶ evaluated, successful, and failed tasks
- ▶ successful and failed runs
- ▶ diagnostic status and total duration

`final_clean_summary.csv`

Consolidated artifacts: timestamp-level records

Forecasts — 208,104 rows

Timestamp-level forecast outputs:

- ▶ dataset, series, origin, model, and repetition
- ▶ seed, forecast step, and timestamp
- ▶ actual, point forecast, and mean forecast
- ▶ quantiles from 0.1 through 0.9

`final_clean_forecasts.csv`

Contexts — 72,682 rows

Historical observations supplied before each cutoff:

- ▶ dataset, series, and rolling origin
- ▶ historical step and timestamp
- ▶ observed value
- ▶ joins with forecasts for full trajectory plots

`final_clean_contexts.csv`

`source_run: keyless | external_keyless | heavier_chronos`